Turning Data into Narrative

SEPTEMBER 11, 2025 | BY PAULINA PRZYSTUPA (EDIT)

Welcome to the Digital Data Stories of the Alexandria Archive Institute and Open Context! In this Data Story Short, we'll talk about what we do with all that data we've created, entered, and analyzed.

The goal of many archaeological projects is to turn data into narrative as a way to create knowledge about the past for people in the present and future. While all humans are storytellers, archaeologists play a special role as storytellers. This is because we tell stories about our own lives and experiences as well as use the data we create and collect to tell stories about people from the past.

In this short, we'll explore how archaeologists create narratives grounded in scientifically-derived findings. We'll begin by explaining a narrative lens and then proceed to writing a short narrative. After that, we'll use some narrative lenses to rewrite our narrative to consider different interpretations of the same data. For this short, we use data from *The Gabii Project* in our examples. Specifically, the data available here. You can use the same data as us, other data from Open Context, or your own data as you work through the following short.

What is a narrative lens?

In an everyday story, the narrative lens is what we use to see and understand the story that we're experiencing. Aka the perspective it's from. As scientists, archaeologists, or other data-driven creators making data narratives, we have to be mindful of our perspective because we're telling a story based on data and our perspective can influence how our audience understands those data. To get a taste for this, we'll look at what makes a narrative "standard" or "hegemonic", at feminist narratives, and at meta-archaeological narratives to explore different narrative lenses or storytelling perspectives.

Hegemonic narratives

A hegemonic narrative in archaeology is the standard, basic story about the past for a particular place and time. It's the baseline approach or perspective for archaeological storytelling. These narratives are useful to understand the archaeological tropes created by archaeologists—and others who use the archaeological record as inspiration—of a particular culture, period, and place. They're not wrong, because sometimes the standard story is the story, but they're not the only one. Common hegemonic narratives in archaeology include stories about the Roman Empire told from the perspective of a soldier or gladiator or stories about Indigenous peoples told from the perspective of a shaman or other spiritual leader.

Feminist narratives

Data-driven stories that use a feminist narrative ask storytellers to see data as contextual. It asks the audience to rethink gender hierarchies or access to material goods or power. While these narratives don't have to have a woman main character or as a central figure, they're more likely to center the perspectives of children, the elderly, disabled people, and others that modern Western society often marginalizes. An example of a feminist narrative lens in archaeology could include stories about the Roman Empire told from the perspective of a child in an occupied territory or archaeological stories by Indigenous peoples told using an existing Indigenous oral tradition that predates colonization.

Meta narratives

A meta-narrative takes a specific experience, such as that of one person, and universalizes it to explain a particular thing. This can be hard to craft without an example so here are few. A meta narrative in archaeology could include: stories about the Roman Empire told from the perspective of a student attending their first field school in Italy or stories about Indigenous North America told from the perspective of an Indigenous archaeologist working in cultural resource management in the United States. Both of these may take the experience of a specific person (sometimes based on the background of the creator of the story) and use it to say more about the field or the period under study.

Writing a narrative

Now that you've learned about a few lenses, you might be ready to create your first data narrative. You can use the following example to guide your thoughts as you create your narrative based on the data you're using for this short. In ours, we underlined certain parts to identify when something in our narrative came from our data.

> *The smell of cooking meat and the stink of other people's bodies was inescapable. Crouching on the uneven floor, Aemilianus shook the die, blowing on it gently. He rattled it out of his hand. It rolled, and rolled, and rolled — there was the six, and then, and then, it tipped, coming to a stop.*
>
> *A two. He couldn't win on a two.*
>
> *Reaching into the pouch affixed to the mail of his armor, he pulled out a few thin bronze coins, flipping them to the winner. Someone called his name, and he looked up. Time to go.*
>
> *He left the die. It wasn't doing him any favors.*

Now, return to the example data set we mentioned at the beginning of this short and see if you can identify what data drove the underlines. After doing that, consider what lens this short narrative uses.

If you can't find the links between this narrative and the data, or want to check your work, explore the following explanations.

1. The first highlighted section refers to three samples of faunal materials (aka animal bones) found in that unit. Cooking seemed like a good way for them to get there.

2. The second highlighted section refers to the shape and layer surface described in the main observations about the unit.
3. The third highlighted section refers to a gaming die listed as a special find in that unit. Someone like *Aemilianus* would probably leave it behind on a bad roll.
4. The fourth highlighted section refers to a bronze coin also listed as a special find in that unit. In our narrative, we decided that the winner forgot to grab a some of their winnings.

And you probably guessed that this is a hegemonic Roman narrative like we might see in film or television, in a novel, on a sign at a site, or in a museum booklet. The clues to look for are that the protagonist is a military man or someone else who often has power in a particular community (or is assumed to have power based on our knowledge of that culture).

Now it's your turn. Pick three pieces of evidence from your data set. Then take five minutes (or so) to create a short one-paragraph story, or draw a few panels of a comic that incorporates your three pieces of data. For this first narrative, use a hegemonic lens appropriate for the where and when of your data. If you're worried about how to get started, consider these ideas:

- What stories have you heard in the past that might make good foundations for this story?
- What kinds of events might bring these three pieces of data together?
- Who might be a typical person to have any artifacts you're incorporating or be in a place like your setting?

It's okay to be outlandish or awkward in the first draft and feel free to write dialogue, draw stick figures or storyboards, or go online to find period-appropriate names to inspire you. Once you've finished, if you'd like to take the chance to analyze your narrative for things like jargon and word choice before jumping into editing and adapting the narrative, check out the "Analyzing your narrative" section of the Digital Data Story we adapted this short from. Otherwise, let's get adapting!

Adapting the narrative

Now that you've created a narrative, it's time to apply different lenses. For each, you'll repeat the process that you used to create your first narrative and reuse your data. The goal with these adaptations is to use different narrative lenses to consider how the same data can create new interpretations. As you look at each lens, you may notice that it's easy to write from that perspective based on the data. If so, identify why the data worked so well (or was really difficult!) with that narrative lens by considering the following questions:

- Are there other pieces of data within the data set that reinforce that perspective or a particular character?
- Are there pieces of data that refute using that specific narrator or story? What data might you need to add to make that specific narrator or story more plausible?
- How could you adapt the data narrative to account for the other data sources or to resolve any contradictions?

If you realize that your original narrative already fits one of the following lenses, swap it out for one that you think might be more "standard" or "hegemonic" for one of your adaptations instead. For hegemonic or standard lenses, think about stories you've already heard, read, or watched about a particular place or what an older generation may see as important about that culture. Then see if you can make your narrative and data work within that space.

Adaptation through a feminist lens

When you create through this lens think: gender, hierarchies, power, and access. You can read our adapted narrative to get an idea of what we mean.

> *It was her turn, but he kept hoarding the die. He hoarded the thin bronze disc, too, and all of the bits of metal they'd scrounged and hidden here, in their secret place.*
>
> *Helvia and Helvius were meant to be minding the goats, but the goats mostly minded themselves. There was time to play, and this falling down room, with its uneven floor, made the perfect place to hide away and pretend.*
>
> *She toed a broken piece of thick-walled pottery and waited for her brother to remember it was her turn.*

This narrative incorporates many of the same artifacts, locations, and context provided by the data in our first narrative, but we replaced the hegemonic narration with actors and goals aligned with feminist archaeological practice. In this case, instead of a military man, an identity associated with access to social and economic power, we challenge unequal power structures by seeing the site from the point of view of a girl whose perspective we experience in comparison to that of a boy of similar age and sociocultural status. For extra practice, see if you can identify the pieces of data from Unit 1158 that went into this version.

As you adapt your narrative to fit this lens, consider the following questions:

- How could I adapt this narrative or reuse my data to challenge archaeological norms of control, power, age, or gender?
- Does this adapted narrative reflect a view of the past through a rarely represented narrator in relation to the tropes I'm aware of for this past community?
- Does the data used as evidence for the narrative still apply and support this new version?

Like your first narrative, it's ok if it's a little awkward or doesn't quite fit as smoothly as you'd like. It's a first draft and the more you work at this kind of thing the better you'll get. So try something and see where this lens takes you.

Rewriting through a meta-archaeological lens

Now that you've tried a couple perspectives, you'll create a meta-archaeological narrative. Creating something using a meta-perspective based on data can be fun. It gives the public (or whatever audience you're creating for) a peek at the realities of archaeological practice (or whatever data-driven discipline you're in), sometimes letting the writer or creator incorporate their own experiences, albeit generalized, into a depiction of the field.

> *It was so hot. It was hot all the time. Standing at the edge of their unit — the ground they had revealed as a small room with an uneven floor, they considered their day's labor, and they sweated.*
>
> *The whole day, spent sweating and dusty, and it would probably end up in the report as just another post- abandonment surface, if it made it in the report at all. Their finds bags had some bones and charcoal, a bronze coin, a few bits of metal, a broken fibula pin, and a compact bone die, the four-side of its matching pips missing. There was a time they'd have had all of these things on their 'to find' bucket list.*
>
> *Time for a Peroni. They signed their name on the context sheet and stretched.*

This narrative adds a bit more data to the mix. Exploring the context sheet for Unit 1158, we cross-referenced the day someone excavated it with historical weather data and discovered that it was a hot day in July outside of Rome. Reviewing even more data, we found out that the same person filled out multiple context sheets that day. By reading their notes, we got an idea of the kinds of things they noticed.

This narrative resonates with many archaeologists (and other field-based workers) experiences but isn't one common in mass media. It's the end of the day, our archaeologist character is tired, they're feeling jaded, yet by sharing that experience through this narrative others can understand what it means to *do* archaeology.

As you apply this lens to adapting your or creating a new narrative, consider these questions:

- Are hegemonic narratives of archaeologists as explorers, adventurers, and collectors (or whatever stereotypes might exist for your field) subverted in this narrative?
- In this new version, is the narrator an archaeologist, field worker, or scientist and how might that influence their understanding of the past in this story?
- Does the data used as evidence for the narrative still apply and do you want to explore some additional data to incorporate into this narrative?

Meta-narratives are often an excellent place to begin if you're struggling to get started with creating a data-driven narrative. That's because you can work from your own experience to get accustomed to crafting it. An easy way to get started is to think of this as writing a personal journal entry or a letter home. Once you do that you can think about how your experience might reflect that of other people or shed some light on archaeology or your discipline. What data would you draw from to color this personal narrative?

The published narrative

Now that you've explored a few short narratives written by the Data Literacy Program team and have hopefully tried creating, adapting, and recreating your own narrative, you can explore the longer official narrative for *The Gabii Project*, which we used in our examples for this short.

Their narrative is called *A Mid-Republican House From Gabii* and draws on more data than we worked with in this exercise. To read it, follow the previous link. As you read parts (or all) of the project's narrative, consider:

- What data you can identify in those parts of the narrative?
- What narrative lens do the authors use? Is there more than one?
- Could you rewrite, recreate, or adapt part of their narrative using one of the lenses we explored in this exercise?

If you'd like to critically examine the official narrative in more detail, for example to help you spot things like jargon, check out this same section in Part Two of the Digital Data Story *Gabbing About Gabii.*

Concluding thoughts

Through this exercise you've learned how to incorporate data into a narrative by using it as inspiration for a short story. Then you applied different narrative lenses to explore how the same data can support telling different stories. Now if you'd like to learn more about transforming archaeological data into narratives, check out the full-length version of this short. Or, if you'd like to extend this practice and make your own, longer data story, play through Part Two of the Digital Data Story *To Disrupt and Delight: Teaching Archaeology Through Narrative* to explore things like audience and other modes of storytelling.

With that, we hope that you've enjoyed turning the data from forms, notes, tables, and images into a few data-driven narratives. We'd love to hear about what you created with this short so please reach out to share your work or any feedback you have on this exercise by emailing us at: datastories@opencontext.org.

References for further reading

These references provide guidance on the pieces linked to in this exercise and are listed in order of appearance in the text. If any links navigate to a broken page, please check the Wayback Machine for an archived copy of the material.

Check out other Digital Data Stories here: https://doi.org/10.6078/M74F1NW0

Find out more about the Alexandria Archive Institute at: https://alexandriaarchive.org/

Explore some data publications over at Open Context: opencontext.org

Get a direct link to *The Gabii Project* here: https://doi.org/10.6078/M7N014GK

Peruse the data we use in our examples (such as the evidence for an uneven floor) at: https://opencontext.org/subjects/0281b2b1-930d-41c6-936f-9add56429998

The evidence for the smell of cooking meat came from these data: https://opencontext.org/subjects/a3b873e9-7fa0-4cd9-be36-ef4b7add5ccd, https://opencontext.org/subjects/81f93fa8-0536-44e4-89c0-6981b48b9241 , and https://opencontext.org/subjects/1e328230-f6c6-4cd7-9f00-da45c2f3cb2e ,

Check out the die found in this unit at: https://opencontext.org/subjects/90c61867-47e9-4046-9f39-f0b251fb15ac

And then a left over bronze coin here: https://opencontext.org/subjects/5b873b2c-b849-4693-92aa-b664ae6fb9d8

Explore Part Two of *Gabbing About Gabii* to learn more about analyzing narratives for jargon and word choice: https://doi.org/10.6078/M7XW4GX4

If you want to check really quickly those try out the readability calculators at: https://readabilityformulas.com/free-readability-formula-tests.php

Read and explore the official narrative for Gabii in *A Mid-Republican House From Gabii* available for free at: https://doi.org/10.3998/mpub.9231782

You might also find *A Mid-Republican House From Gabii* under the traditional academic citation "Opitz et al. 2018," which is referred to as the "author date" system for citation.

Use our handy game-based decision tree to expand your narrative here: https://doi.org/10.6078/M7Z31WSJ

Then read the Digital Data Story it's part of (aka *To Disrupt and Delight: Teaching Archaeology Through Narrative*) here: https://doi.org/10.6078/M7X928FR

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